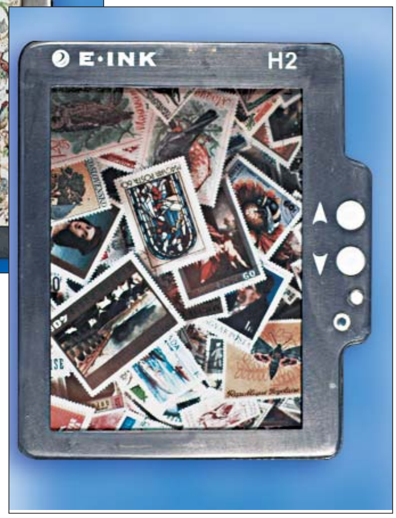

Shown here is the new colour display that has been produced by E-Ink and Toppan Printing. The image above shows you the device's capability of displaying text, while the image on the right shows off the decent image resolution offered—all in 12-bit colour of course!

E-Ink And Toppan Printing's Colour Paper Display

We foretold the coming of this product in *Gadgetopia* in the July 2005 issue, and just six months later, here it is!

This is a colour paper display screen that was developed using E-Ink's display technology and Toppan Printing's flat panel colour filters. Incidentally, Toppan Printing is the world's largest supplier of colour filter arrays for flat panel monitors.

The display offers a sharpness of 83 pixels per inch at a resolution of 400 x 300, and achieves 12-bit colour! The custom-made colour filter array, provided by Toppan, uses high-brightness colour (RGBW) to maintain paper-quality contrast. Basically this means that the white is as good as paper, and the black and other colours are shown up in high contrast to give the display a proper magazine page look!

The screen is six inches in diameter, which is the size of an ordinary paperback novel. This step forward by E-Ink and Toppan now makes the dream of colour paper displays in mobile devices an achievable reality!

Since the display uses minimal power and has a readability angle and contrast on par with paper, it's already got conventional mobile displays beat. The colour display reportedly uses as much as 100 times less power than a conventional LED, which means that battery life on existing mobile products can easily double if this technology is used—not to mention a sizeable reduction in weight!

Perhaps the first adopters of this technology will be everyday devices such as electronic sign boards, digital cameras (for the preview screens), ATMs, GPS devices, and kiosks, thanks to the amazing readability under direct sunlight. Later, we might see the display being used in mobile phones, PDAs, laptops, and other mobile devices that need to save on battery power. The product is said to be able to hit production lines soon—by the end of 2006.

The Return Of Paper

If, or rather, when, these displays are successfully made flexible, perhaps all of us will be able to enjoy gadgets that don't keep running out of battery life, and have screens we can actually read easily in the daytime.

Although we normally cover technologies of the near future in this section, this time we focussed on paper displays, which were first invented in the '70s. The reason for this is: within the next two years, we see paper displays being used a lot more, and becoming a part of everyday life, thanks to their reduced manufacturing costs and battery use.

Since every development that's happened in the past few years has revolved around mobility and the ease of information flow, paper displays seem to fit right into the mould for the future. They're cheaper, less fragile, offer better readability, are lighter, and use less battery power. Now, if only manufacturers and researchers can find a way to rival LCD and CRT refresh rates, we're set to see the return of the paper era, albeit in the new form of e-paper!

Would you be interested in buying any of the products talked about in this article, as and when they're available? Does the idea of an interactive *Digit* printed on e-paper excite you? Write in and tell us—we'd love to hear your take on the subject. ⊠

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